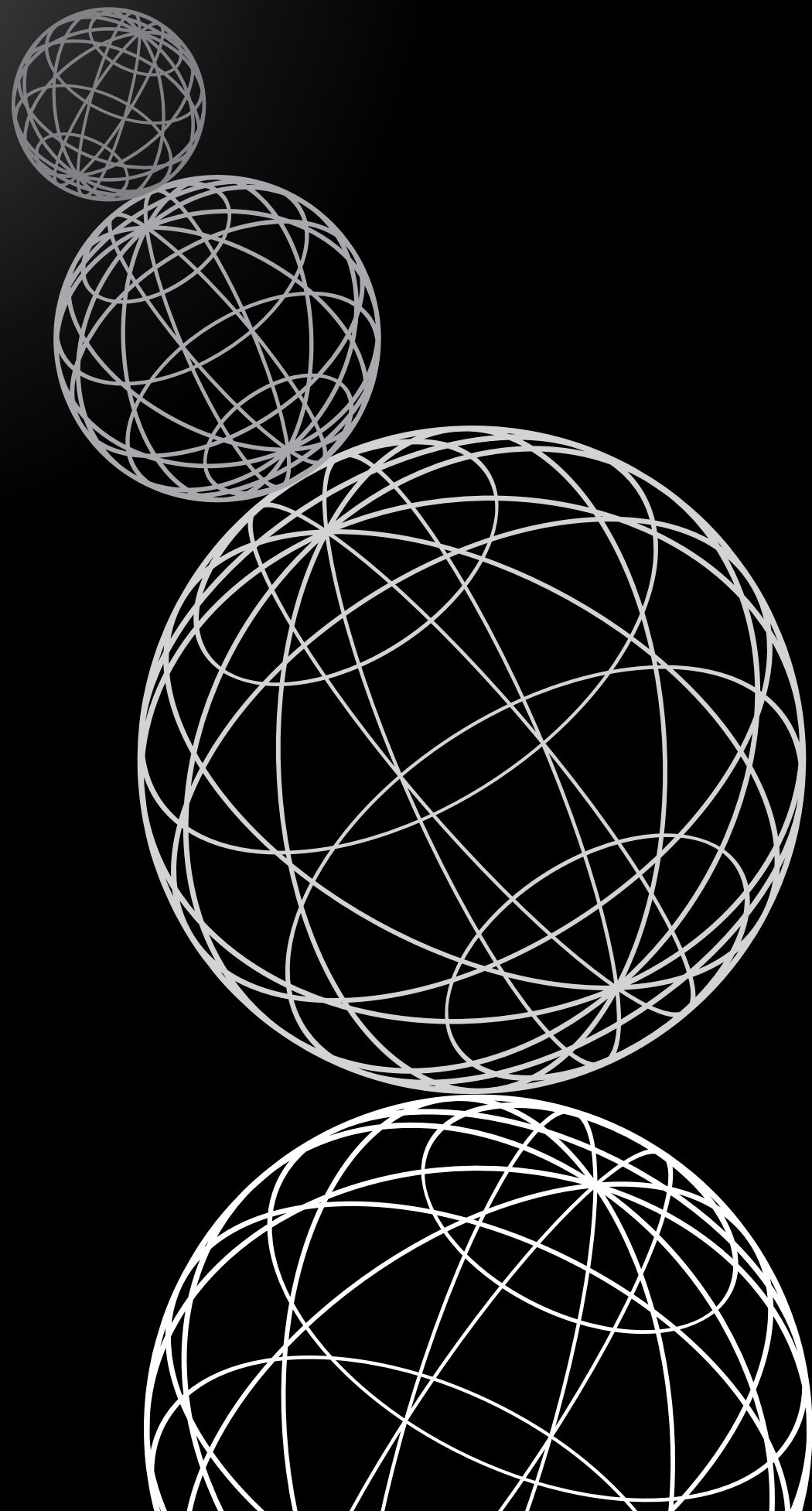




CRAPY RUN!

GABRIEL ASSIS
GABRIEL SICILIANI



INTRODUÇÃO

Utilizando bibliotecas e assets do Python e Godot respectivamente, implementamos um bot que aprende a jogar um jogo que criamos no Godot.

METODOLOGIA

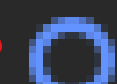


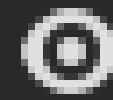
GODOT
Game engine



▼  player

 CollisionShape2D

 AIController2D



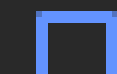
```
func get_obs() -> Dictionary:
>| var obstacle_pos : Vector2
>| if _player.obstacle:
>| >| obstacle_pos = to_local(_player.obstacle.global_position)
>| else:
>| >| obstacle_pos = Vector2.ZERO
>| var playerPos : Vector2 = to_local(_player.global_position)
>| var playerSize : Rect2 = _player.shape.get_shape().get_rect()
>| var obs = [obstacle_pos.x, obstacle_pos.y, playerPos.x, playerPos.y, playerSize.size.x, playerSize.size.y]
>|
>| return {"obs" : obs}

func get_reward() -> float:
>| return reward

func get_action_space() -> Dictionary:
>| return {
>| >| "jump" : {"size" : 3, "action_type" : "discrete"},}

func set_action(action) -> void:
>| match int(action.jump):
>| >| 0:
>| >| >| get_parent().jump = 0
>| >| 1:
>| >| >| get_parent().jump = 1
>| >| 2:
>| >| >| get_parent().jump = 2
```

▼  goal

 CollisionShape2D

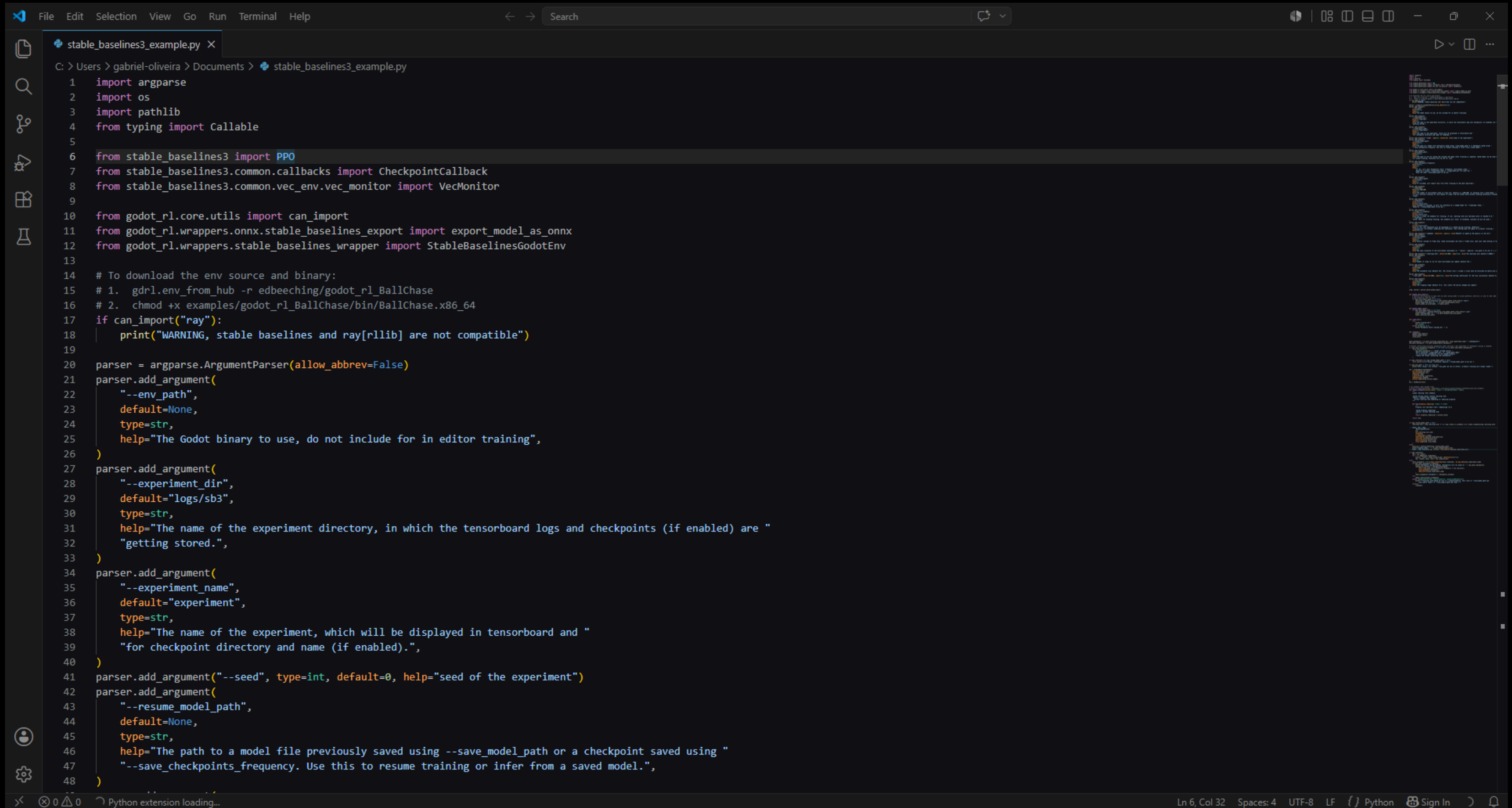


```
▼ func _on_body_entered(body: Node2D) -> void:
▼ >| if body.is_in_group("obstacle"):
>| >| #print(body.name + " Entered " + name)
>| >| player.ai_controller.reward += 10
>| >| points += 1
>| >| pointLabel.text = str(points)
>| >| createObstacle()
>| >| await get_tree().create_timer(0.75).timeout
>| >| body.queue_free()

▼ func createObstacle():
>| >| var newObs : RigidBody2D = obstacleList.pick_random().instantiate()
▼ >| >| match newObs.name:
>| >| >| "SMALL":
>| >| >| >| newObs.position = small.position
▼ >| >| >| "BIG":
>| >| >| >| newObs.position = big.position
>| >| get_parent().call_deferred("add_child", newObs)
>| >| player.newObs(newObs)

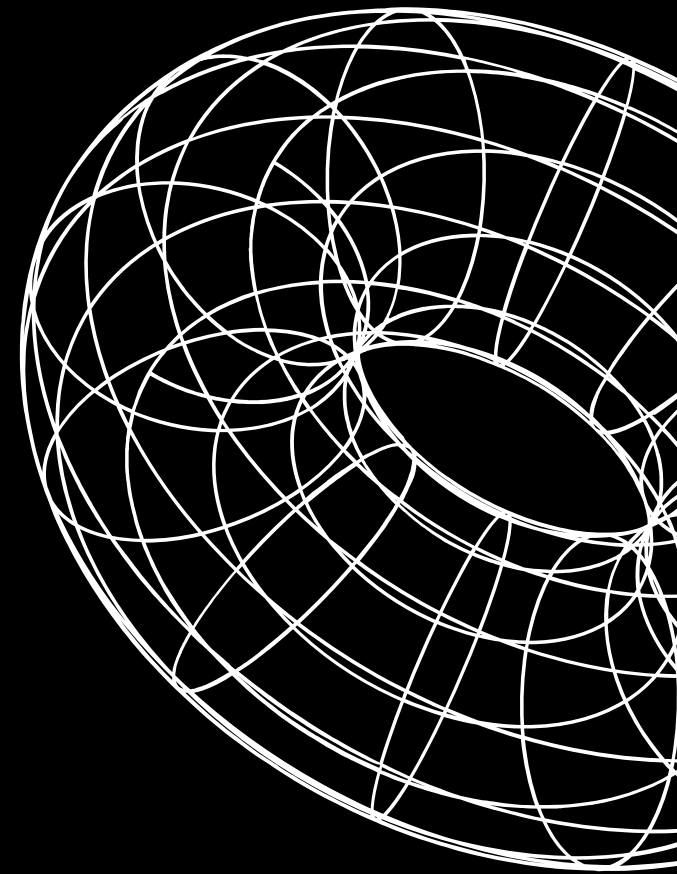
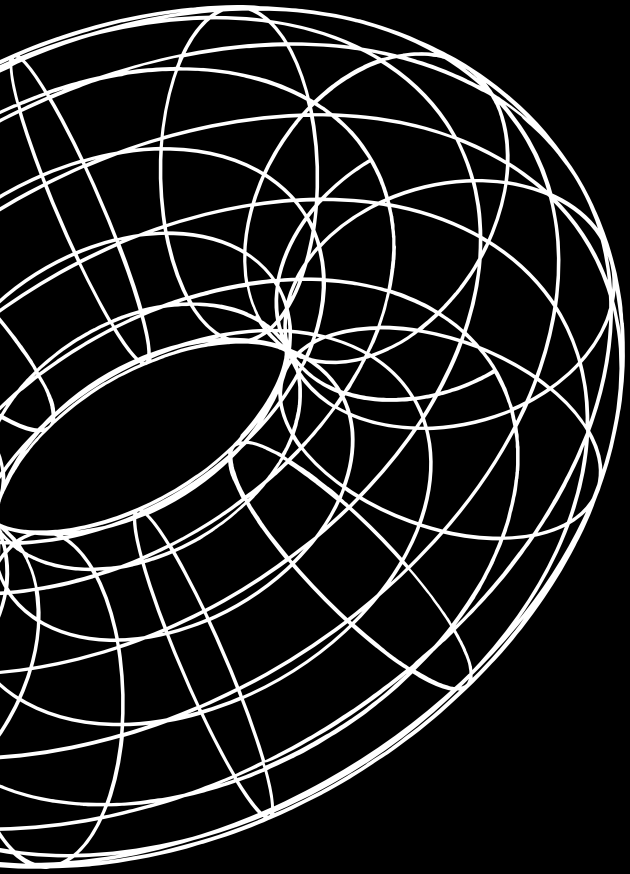
▼ func reset():
>| points = 0
>| pointLabel.text = str(points)
>| createObstacle()
```

https://github.com/edbeeching/godot_rl_agents/blob/main/examples/stable_baselines3_example.py



```
1 import argparse
2 import os
3 import pathlib
4 from typing import Callable
5
6 from stable_baselines3 import PPO
7 from stable_baselines3.common.callbacks import CheckpointCallback
8 from stable_baselines3.common.vec_env.vec_monitor import VecMonitor
9
10 from godot_rl.core.utils import can_import
11 from godot_rl.wrappers.onnx.stable_baselines_export import export_model_as_onnx
12 from godot_rl.wrappers.stable_baselines_wrapper import StableBaselinesGodotEnv
13
14 # To download the env source and binary:
15 # 1. gdr1.env_from_hub -r edbeeching/godot_rl_BallChase
16 # 2. chmod +x examples/godot_rl_BallChase/bin/BallChase.x86_64
17 if can_import("ray"):
18     print("WARNING, stable baselines and ray[rllib] are not compatible")
19
20 parser = argparse.ArgumentParser(allow_abbrev=False)
21 parser.add_argument(
22     "--env_path",
23     default=None,
24     type=str,
25     help="The Godot binary to use, do not include for in editor training",
26 )
27 parser.add_argument(
28     "--experiment_dir",
29     default="logs/sb3",
30     type=str,
31     help="The name of the experiment directory, in which the tensorboard logs and checkpoints (if enabled) are "
32     "getting stored.",
33 )
34 parser.add_argument(
35     "--experiment_name",
36     default="experiment",
37     type=str,
38     help="The name of the experiment, which will be displayed in tensorboard and "
39     "for checkpoint directory and name (if enabled).",
40 )
41 parser.add_argument("--seed", type=int, default=0, help="seed of the experiment")
42 parser.add_argument(
43     "--resume_model_path",
44     default=None,
45     type=str,
46     help="The path to a model file previously saved using --save_model_path or a checkpoint saved using "
47     "--save_checkpoints_frequency. Use this to resume training or infer from a saved model.",
48 )
```


RESULTADOS



CONSIDERAÇÕES FINAIS

- Facilidade de implementação na parte do GODOT;
- Problemas com instalação de bibliotecas do python;

```
error: subprocess-exited-with-error

x Preparing metadata (pyproject.toml) did not run successfully.
  exit code: 1
  -> [21 lines of output]
    + C:\Users\gabriel-oliveira\AppData\Local\Programs\Python\Python314\python.exe C:\Users\gabriel-oliveira\AppData\Local\Temp\pip-install-_e2iftzr\numpy_58e60b216b134d08ad245b3068879248\vendored-meson\meson\meson.py setup C:\Users\gabriel-oliveira\AppData\Local\Temp\pip-install-_e2iftzr\numpy_58e60b216b134d08ad245b3068879248 C:\Users\gabriel-oliveira\AppData\Local\Temp\pip-install-_e2iftzr\numpy_58e60b216b134d08ad245b3068879248\mesonpy-tykiew11 -Dbuildtype=release -Db_ndebug-if-release -Db_vsctrl-md --native-file=C:\Users\gabriel-oliveira\AppData\Local\Temp\pip-install-_e2iftzr\numpy_58e60b216b134d08ad245b3068879248\mesonpy-tykiew11\meson-python-native-file.ini
    The Meson build system
    Version: 1.2.99
    Source dir: C:\Users\gabriel-oliveira\AppData\Local\Temp\pip-install-_e2iftzr\numpy_58e60b216b134d08ad245b3068879248
    Build dir: C:\Users\gabriel-oliveira\AppData\Local\Temp\pip-install-_e2iftzr\numpy_58e60b216b134d08ad245b3068879248\mesonpy-tykiew11
    Build type: native build
    Project name: NumPy
    Project version: 1.26.4
    WARNING: Failed to activate VS environment: Could not parse vswhere.exe output

    ..\meson.build:1:0: ERROR: Unknown compiler(s): [['icl'], ['cl'], ['cc'], ['gcc'], ['clang'], ['clang-cl'], ['pgcc']]
    The following exception(s) were encountered:
    Running `icl ""` gave "[WinError 2] O sistema não pode encontrar o arquivo especificado"
    Running `cl /?` gave "[WinError 2] O sistema não pode encontrar o arquivo especificado"
    Running `cc --version` gave "[WinError 2] O sistema não pode encontrar o arquivo especificado"
    Running `gcc --version` gave "[WinError 2] O sistema não pode encontrar o arquivo especificado"
    Running `clang --version` gave "[WinError 2] O sistema não pode encontrar o arquivo especificado"
    Running `clang-cl /?` gave "[WinError 2] O sistema não pode encontrar o arquivo especificado"
    Running `pgcc --version` gave "[WinError 2] O sistema não pode encontrar o arquivo especificado"

    A full log can be found at C:\Users\gabriel-oliveira\AppData\Local\Temp\pip-install-_e2iftzr\numpy_58e60b216b134d08ad245b3068879248\mesonpy-tykiew11\meson-logs\meson-log.txt
    [end of output]
```

REFERÊNCIAS

<https://huggingface.co/learn/deep-rl-course/unitbonus3/godotrl>

https://github.com/edbeeching/godot_rl_agents

https://www.youtube.com/watch?v=f8arMv_rtUU

